



What can an agent do?

When we create an agent it normally has a number of different actions that it can perform, or ways that it can interact with the environment. However, this does not mean that each action can be performed in all circumstance. For example, if the agent has an action to 'pick up the cat' then this is only appropriate if the cat is light enough to be lifted by the agent, the cat wants to be lifted up or there is even a cat around.

What this means is that actions have preconditions associated with them, a precondition defines the possible situations in which they can be applied. If we go back to our example then an agent can lift a cat when:

- The cat is happy to be lifted up
- The cat is light enough for the agent to lift
- There is a cat present.

These preconditions define the situations where the actions can be applied.

Understanding these preconditions is a key element of agent design and relates to agent architectures, we will focus on this more in future lecture casts but for now we will simply say that agent architectures can help the agent decide which of its actions should be performed to best meet its goals.